Golfit

Objective

The objective of this project is to build a multiplayer real-time golf game, where players compete to finish the course in the fewest strokes possible.

Unlike traditional golf, the environment is interactive. Players can trigger environmental changes that allow competitive advantages or disadvantages, such as moving bridges or closing gates. This adds more strategic play to golfit and allows more player-to-player interactions.

The project is successful if there is a functional golf game with scoring, game chat, realistic ball physics and a complete client-server network for the game.

Game concept

The game concept is outlined clearly in Game_Concept.pdf.

Highlights :

- Real-time golf instead of turn-based.
- Focus on strategic gameplay and social interaction.
- Interactive map elements.

Organization

Name	**Responsibility**
Demian	Graphics and Game Logic
Tamer	Physics
Constantin	Graphics and Game Logic
Anton	Networking

- Regular updates on progress
- Diary entry after each work
- Group call when required

Communication tools are Whatsapp and Discord.

Tasks will be weekly, exact dates do not matter. The last week before milestones will be used to polish the game and make it presentable.

Networking

All requirements and the protocol is outlined in requirementAnalysis.md and networkProtocol.md .

Since the game is real-time, techniques such as client-side prediction are required to keep the game smooth. Security measures are outside of this project's scope.

Packages

Graphics : JavaFX

Testing : Junit

Logging : Log4j 2

Networking: Java Sockets

Timeline

The complete timeline (including tasks) is visualized in Timeline.pdf. The first weeks focus on developing the groundwork and including basic mechanics such as the golf hole, ball kick and creating maps. However the order does not matter and can be swapped.

Key dates and milestones:

Week/Date	What	Who	Comments
6th March 11 :10	Milestone 1	All	Details : Description and game rules, software requirements, functionalities of client-server, project organisation
18th March	Milestone 2 (Exercise)	All	Details : implementation of system with simple socket connection
26th March	Milestone 3	All	Multiple Choice
13th April	Exercise Group	All	Show process
16th April	Milestone 4	All	Core components without rule check and GUI, GUI for lobby and chat
27th April	Milestone 5 (Exercise)	All	First GUI version for game, complete game rules Goal : first functioning client/server with GUIM
11th May	Exercise	All	Show process

21.05-29.05	Milestone 6	All	Present whole project, live demo, test network beforehand, screenshot, QA measurements, lessons learned

Notes :

- Commit all files and documentation in Gitlab
- Focus on meeting milestone requirements